



ICCIT 2025

28th INTERNATIONAL CONFERENCE ON COMPUTER AND INFORMATION TECHNOLOGY



Long Beach Hotel
Cox's Bazar, Bangladesh



DECEMBER 19 - 21, 2025



<http://iccit.org.bd/2025/>

Paper ID: 562

A Transformer-Based Framework for Multi-Agent Trajectory Prediction with Region-Aware Representation and Temporal Encoding

Spandan Basu, Barshon Sen, Tanvir Anjom Siddique, Kanchon Gharami

Rajshahi University of Engineering & Technology, Rajshahi University of Engineering & Technology,
Rajshahi University of Engineering & Technology, Embry-Riddle Aeronautical University



Presented By: Spandan Basu

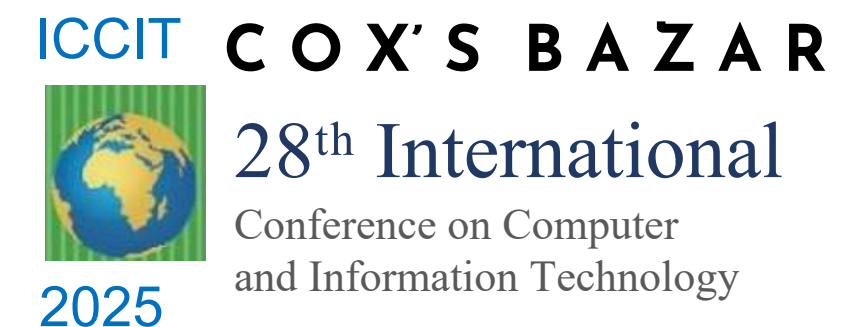


Table of Contents



1
Introduction

2
Challenges

3
**Research
Objectives**

4
**Literature
Review**

5
**Dataset
Description**

6
Methodology

7
**Result
Analysis**

8
Conclusion

9
References



Introduction

Autonomous Driving

- Self-Driving Technology.
- Driverless Vehicle
- Intelligent route observation

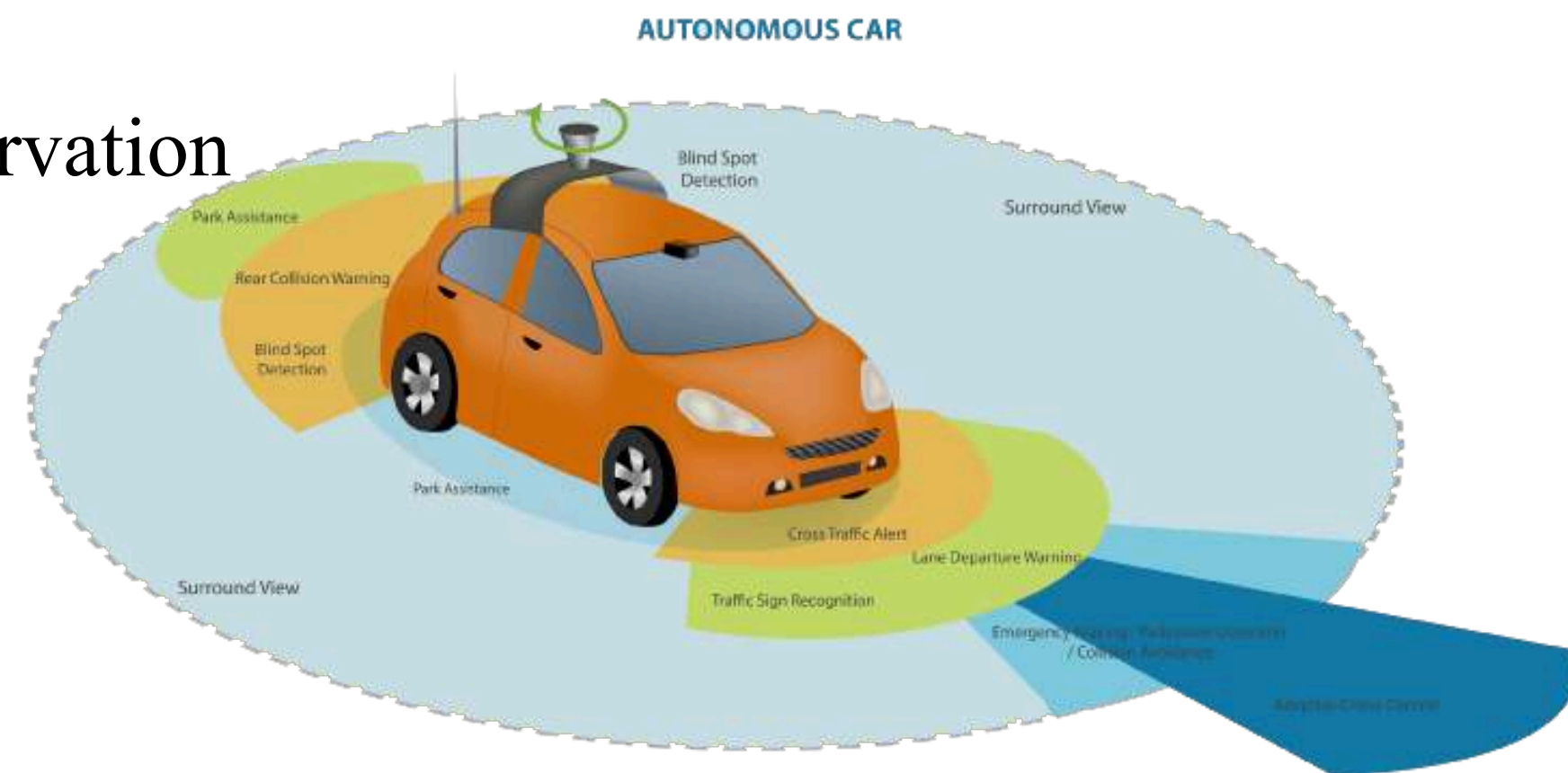


Fig 1: Autonomous Vehicle's Components and placements[1]



Introduction(cont'd)

Necessity of Autonomous Vehicle(AV)

- Safety on roads. [2]
- More Disciplined.
- Highly aligned with electric vehicles.
- Manage Human Workload.



Road
Safety



More
Discipline



EV
Alignment

Manage
Human
Workload

Challenges

Key points of Autonomous Vehicle



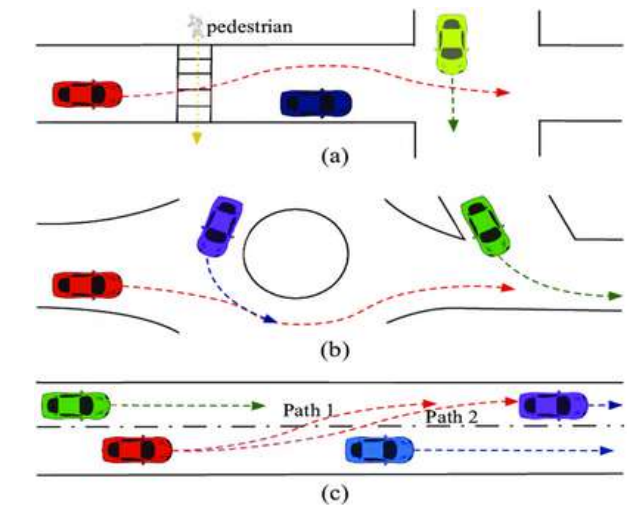
Perception



Mapping



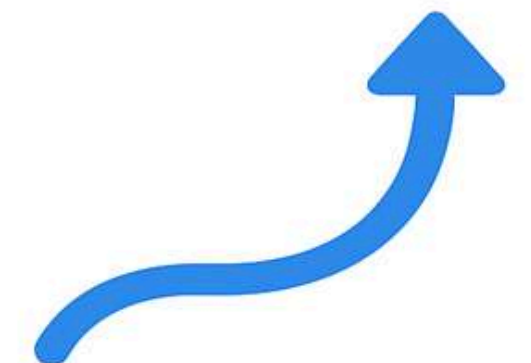
Destination
Planning



Trajectory
Prediction

Research Objectives

- Transport issues identification.
- Safe riding.
- Coordinate management.
- Efficient Prediction



Literature Review

Paper-1

TraPHic: Trajectory Prediction in Dense and Heterogeneous Traffic
Using Weighted Interactions.[3]



Contribution

- Handle dense and heterogeneous traffic.
- New interaction modeling approach.
- Up to 30% improvement in prediction accuracy.



Experimental Result

- $RMSE_1 = 0.23$
- $RMSE_3 = 1.52$



Limitation

- Relies on high-quality annotated data.
- High Computational Complexity.
- Limited to Short-Term Predictions.



Literature Review (Cont'd)

Paper-2

BAT: Behavior-Aware Human-Like Trajectory Prediction for Autonomous Driving[4]



Contribution

- Behavior-aware modeling.
- Priority-aware attention.
- High performance with reduced training data.



Experimental Result

- $RMSE_1 = 0.42$
- $RMSE_3 = 1.69$



Limitation

- Limited real-world validation.
- Excludes other road users.
- Long training due to multi-component architecture.



Literature Review (Cont'd)

Paper-3

Convolutional Social Pooling for Vehicle Trajectory Prediction[5]



Contribution

- Social LSTMs with convolution + max-pooling.
- Multimodal distribution over six high-level classes.



Experimental Result

- $RMSE_1 = 0.62$
- $RMSE_3 = 1.55$



Limitation

- Limited Generalizability.
- Lack of Explainability.
- Limited test scenarios.

Literature Review (Cont'd)

Paper-4

Dual Transformer Based Prediction for Lane Change Intentions and Trajectories in Mixed Traffic Environment[6]



Contribution

- Higher accuracy.
- Leveraged multi-head attention.
- Dual Transformer join prediction of lane change intentions and trajectories.



Experimental Result

- $RMSE_1 = 0.67$
- $RMSE_3 = 1.21$



Limitation

- Prioritize only smoothness.
- Large-scale deployment is not addressed.
- Limited scalability of the approach for real-world.



Dataset Description

Argoverse 1.1 (2019)[7]

- 3D Mapping
- Motion Forecasting
- 64-beam lidar, stereo imagery, GPS/IMU
- 15 annotated object classes

Table 1: Basic information about dataset

Property	Values
Number of Cities	2
City Names	Pittsburg, Miami
Coordinate System	Local Scene Coordinates
HD map	Available



—

Dataset Description (Cont'd)

Argoverse 1.1
(2019)

Table 2: Full Dataset Specifications

Property	Values
Total no. of files	324,000
No of files after preprocessing	170,090
No of Training Files	136,720
No of Test Files	No of Test Files
Avg Seq duration	0.9 sec

Table 3: Vehicle Specifications

Object Type	Approximate Number	Percentage
Agent Vehicle (Ego Vehicle)	324,000	~ 10%
Vehicle (Non agent)	1,800,000	~55%
Pedestrian	450,000	~30%
Others/ Unknown	50,000	~5%

Methodology

System Overview

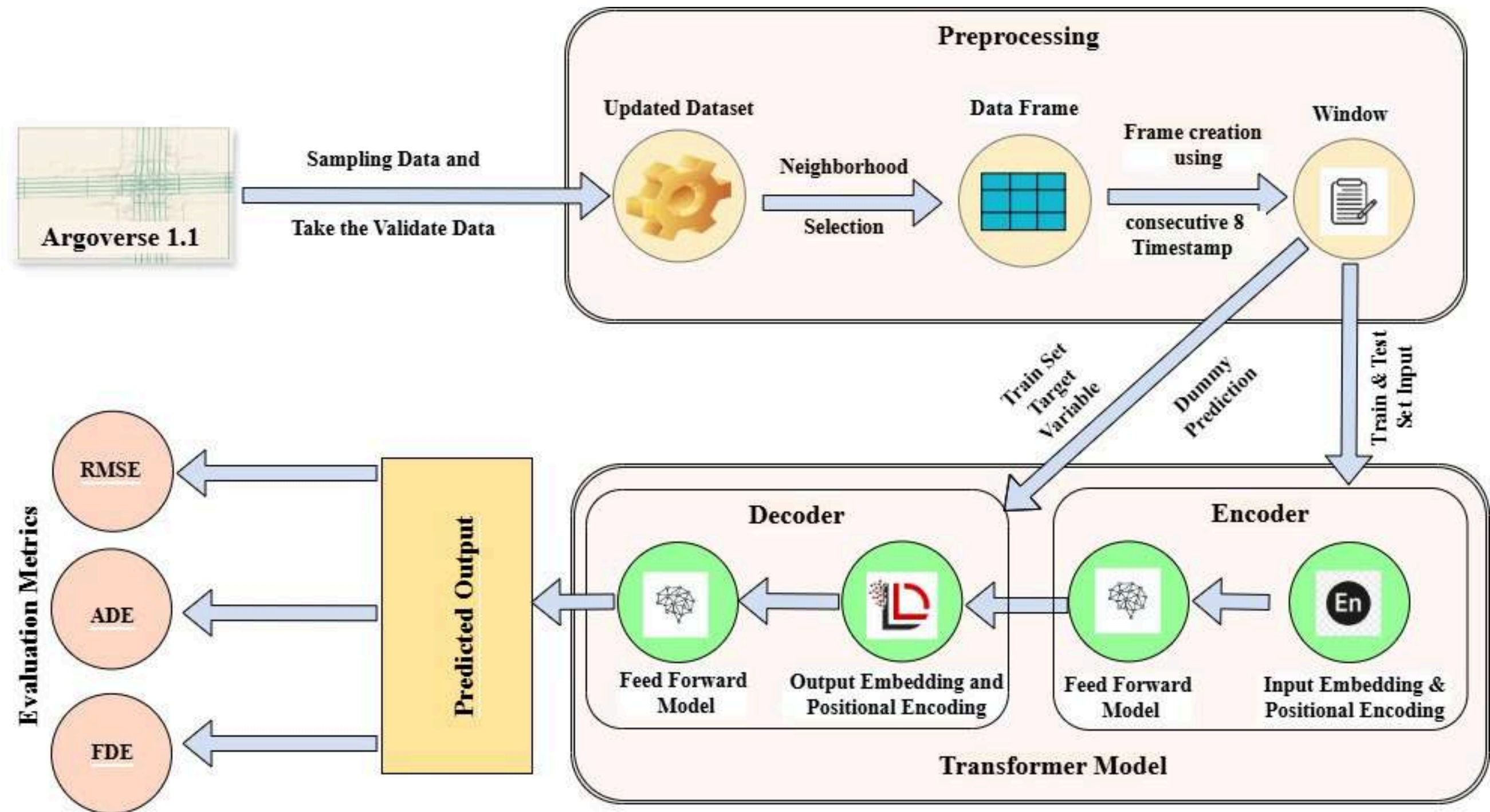


Fig-02: Block Diagram of our work approach

Methodology (Cont'd)

Preprocessing



- **Input agent sequence:**

$$\mathbf{X}^{(i)} = \begin{bmatrix} \mathbf{x}_{\text{main}}^{(i)} \\ \mathbf{x}_{\text{main}}^{(i+1)} \\ \vdots \\ \mathbf{x}_{\text{main}}^{(i+s-1)} \end{bmatrix} \in \mathbb{R}^{s \times 2}$$

- **Obstacle features:**

$$\mathbf{o}^{(i+s-1)} \in \mathbb{R}^{16}$$

- **Target:**

$$\mathbf{y}^{(i)} = \mathbf{x}_{\text{main}}^{(i+s)} \in \mathbb{R}^{1 \times 2}$$

Normalization:

$$\mathbf{X}_{\text{norm}} = \frac{x - \mu}{\sigma} \quad \text{----- (1)}$$

μ = Mean value of the main agent's positions

σ = Standard Deviation of the main agent's positions

$\mathbf{x}^{(i)}$ = Agent vehicle position for i^{th} sequence

$\mathbf{O}^{(i+s-1)}$ = Obstacle position for $(s-1)^{\text{th}}$ sequence.

$\mathbf{y}^{(i)}$ = target output for $(s-1)^{\text{th}}$ sequence.

$\mathbf{X}_{\text{norm}}$ = Normalized value of agent

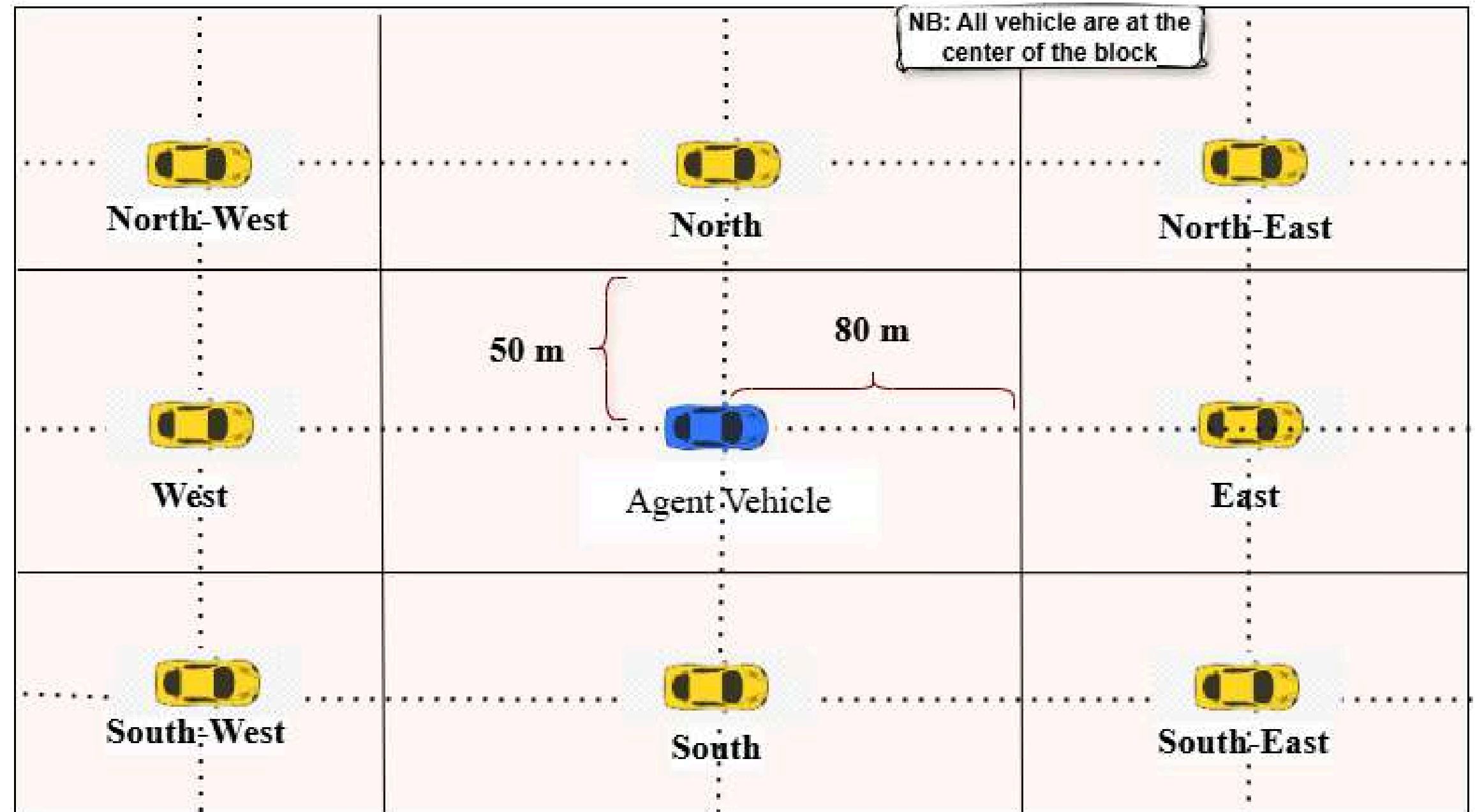
[illegible]

Fig-03: Block Diagram of vehicle's region management

Methodology (Cont'd)

Transformer Model

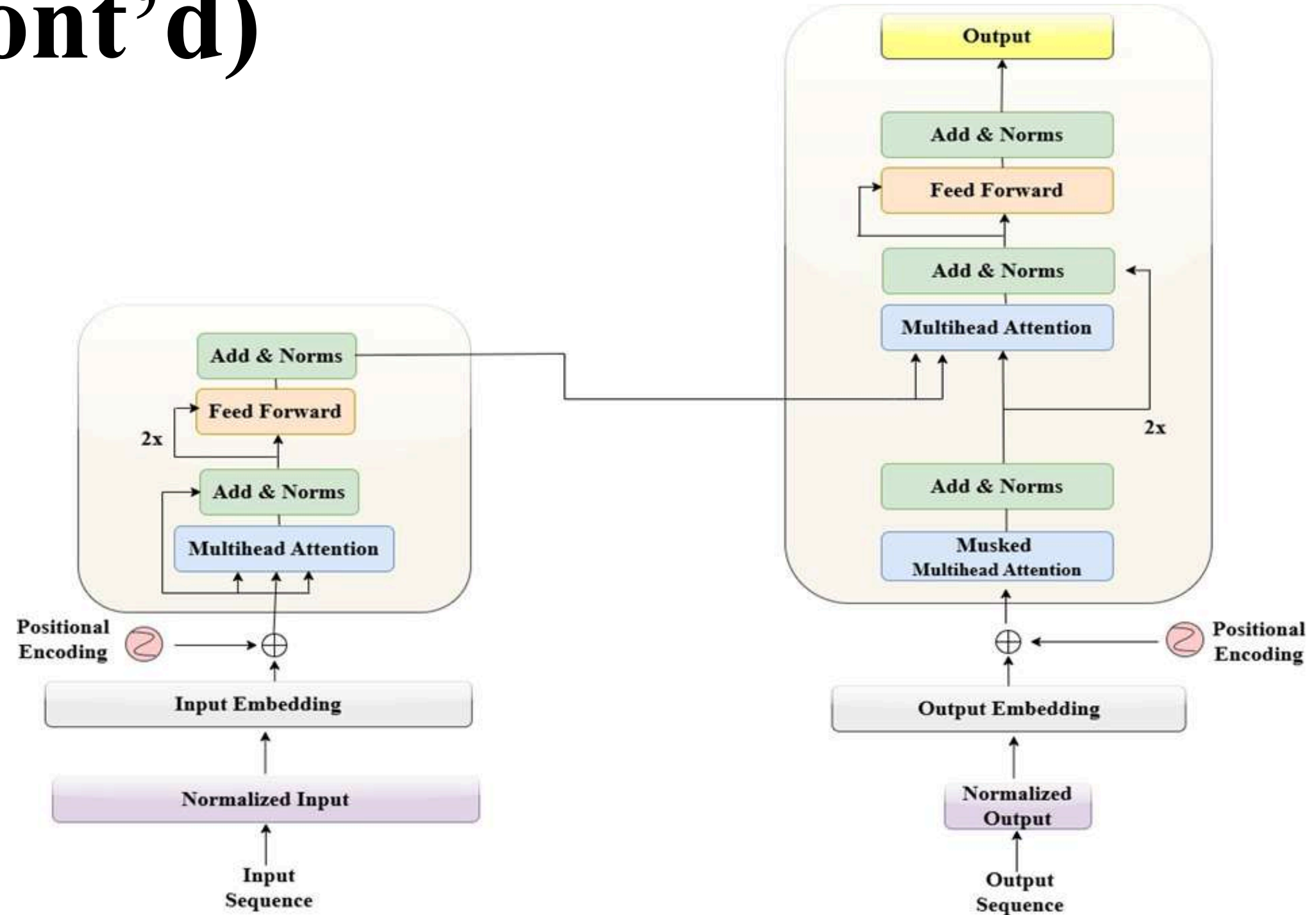


Fig-04: Block Diagram of vehicle's region management

Methodology (Cont'd)

Transformer Model

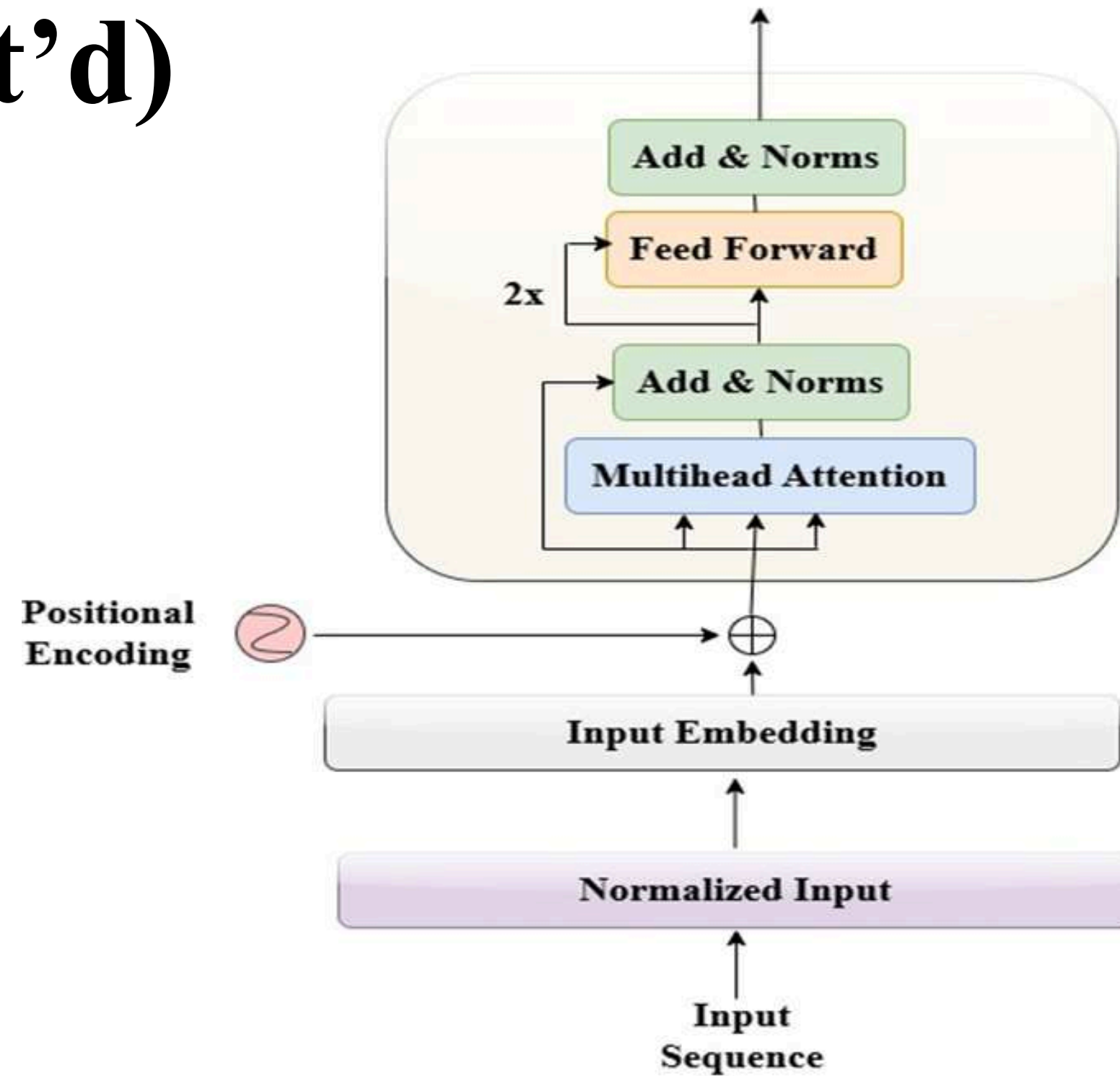


Fig-05: Encoder Diagram

Methodology (Cont'd)

Transformer Model

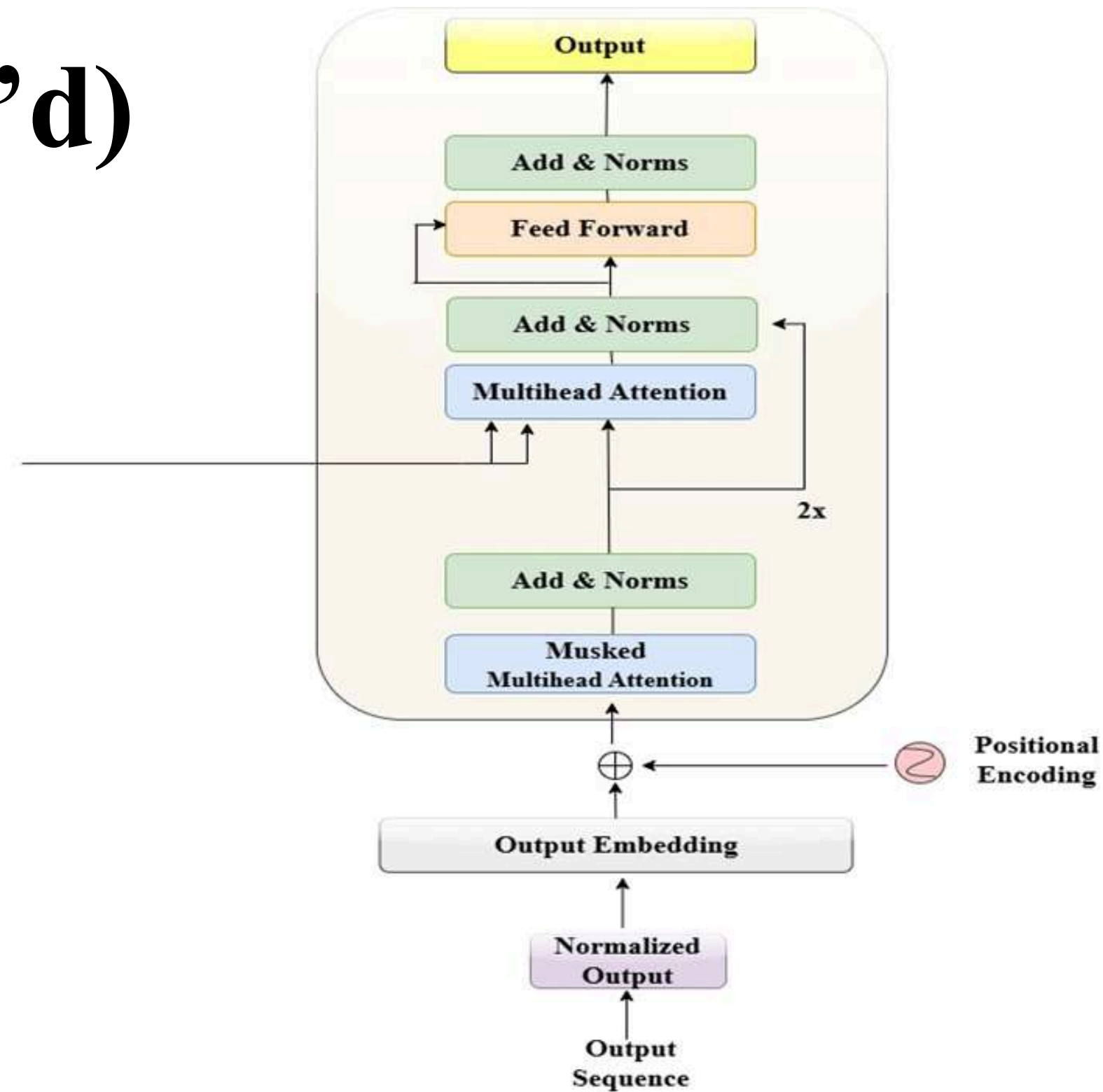


Fig-06: Decoder Diagram

Methodology (Cont'd)

Positional Encoding



- Past position Encoding
- Attention-based motion capture.
- Prediction horizon generalization

$$\mathbf{P}_{i,2k} = \sin\left(\frac{5i}{1000^{2k/d_{model}}}\right) \text{-----}(2)$$

$$\mathbf{P}_{i,2k+1} = \cos\left(\frac{5i}{1000^{2k/d_{model}}}\right) \text{-----}(3)$$

- i = Individual timestamp ,
- k = the frame/window which currently processed,
- d_{model} = dimensionality model embeddings

Methodology (Cont'd)

Attention Mechanism



Single Head: Query Q, Keys K and Value $V \in \mathbb{R}^{L \times B \times d_k}$

$$\text{Atten}(Q, K, V) = \frac{QK^T}{\sqrt{d_k}} V \text{-----(4)}$$

Multi Head:

$$\text{Split model } d_{model} = h d_k \text{-----(5)}$$

$$head_i = \frac{(QW_i^Q)(KW_i^K)}{\sqrt{d_k}} (VW_i^V) \in \mathbb{R}^{L \times B \times d_k} \text{-----(6)}$$

$$\text{Multi_Head}(Q, K, V) = \text{concat}(head_1, head_2, head_3, \dots, head_h) W^O \text{-----(7)}$$

Methodology (Cont'd)

Loss Curve



Fig-07: Generated Loss Curve during training and validation.

Result Analysis

Evaluation Metrics



Table 5: ADE and FDE values at different epochs

Number of Epochs	ADE	FDE
1	0.9065	0.9251
5	0.7077	0.7606
10	0.6378	0.6989
15	0.6243	0.6957
20	0.6268	0.6527
25	0.5803	0.5971

Result Analysis(Cont'd)

Predicted Trajectory of Implemented Model

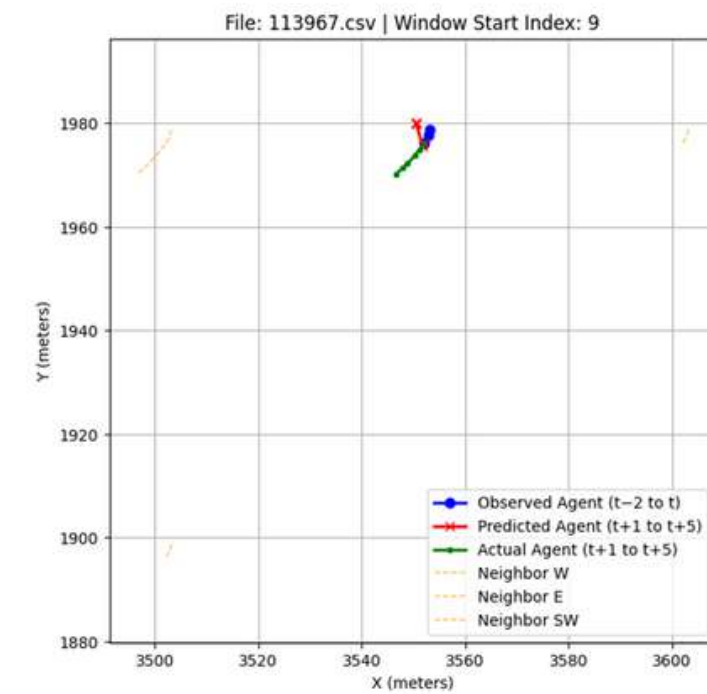
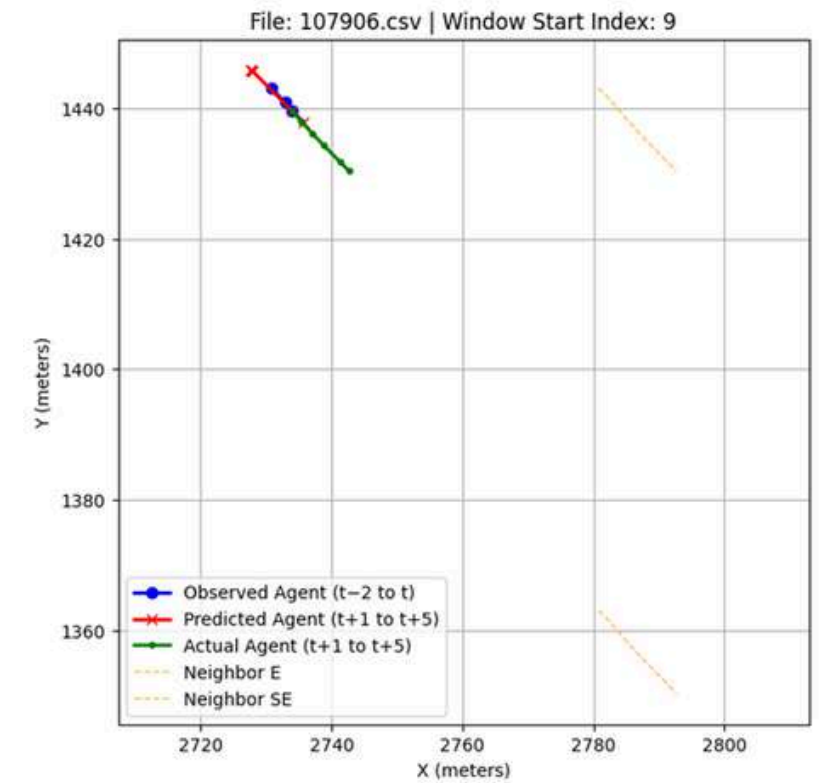
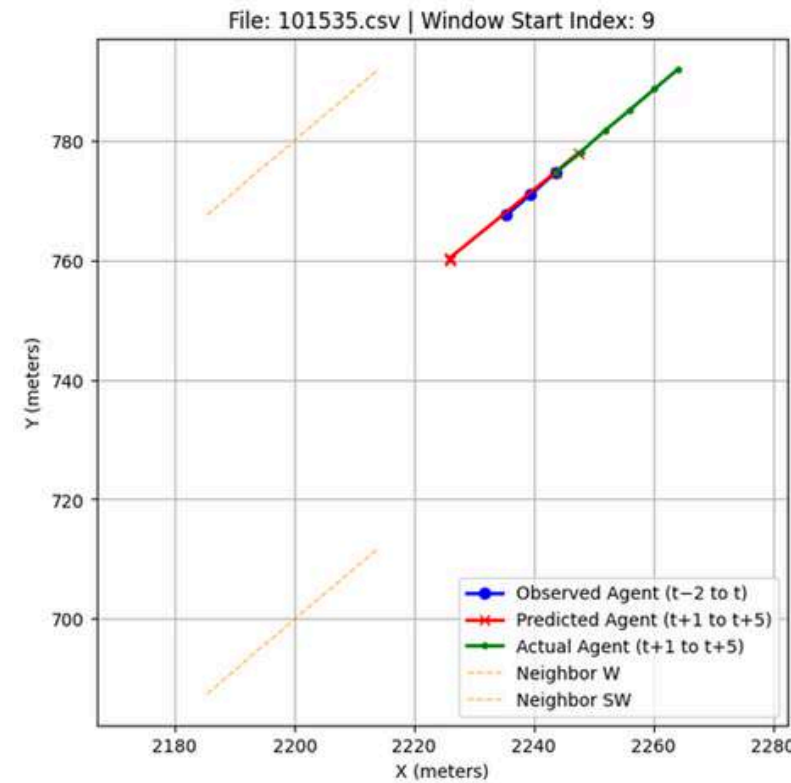


Fig-08: Predicted Trajectories after 1st Epoch

Result Analysis(Cont'd)

Predicted Trajectory of Implemented Model

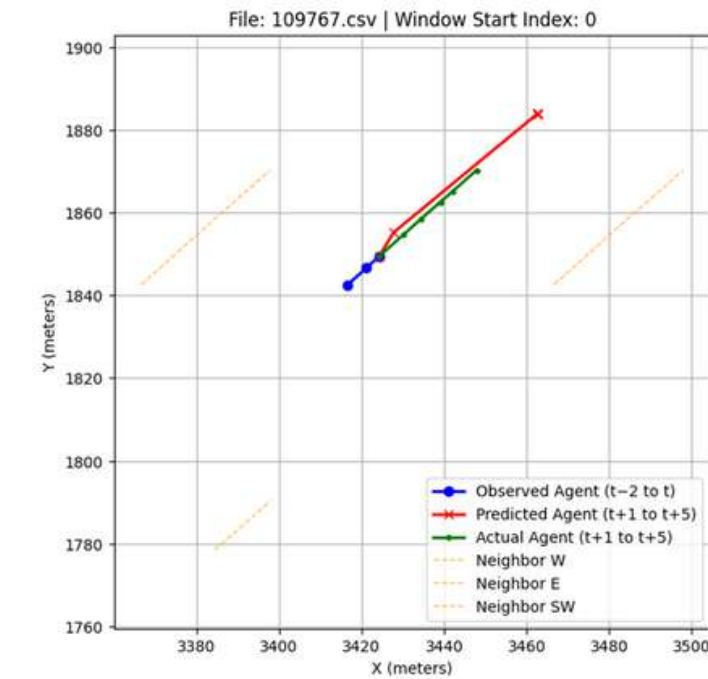
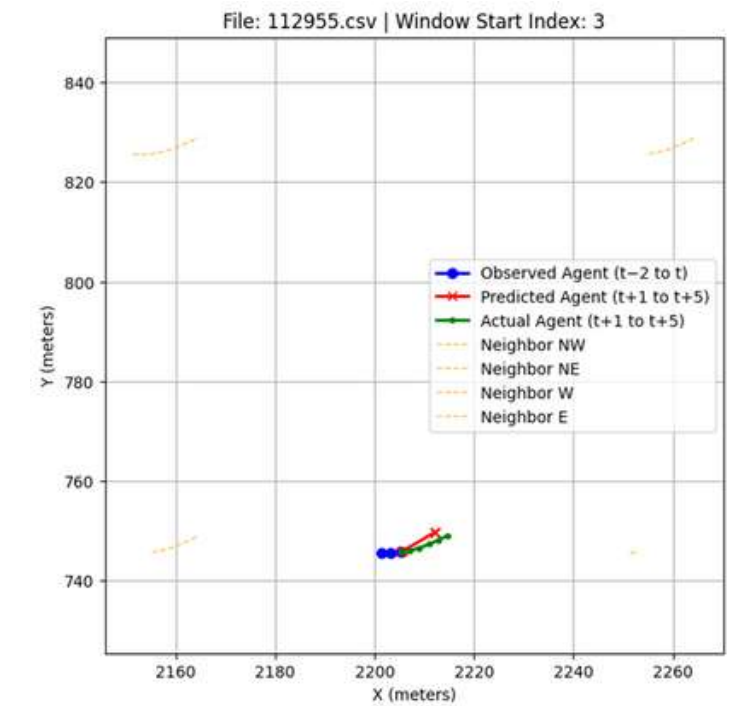
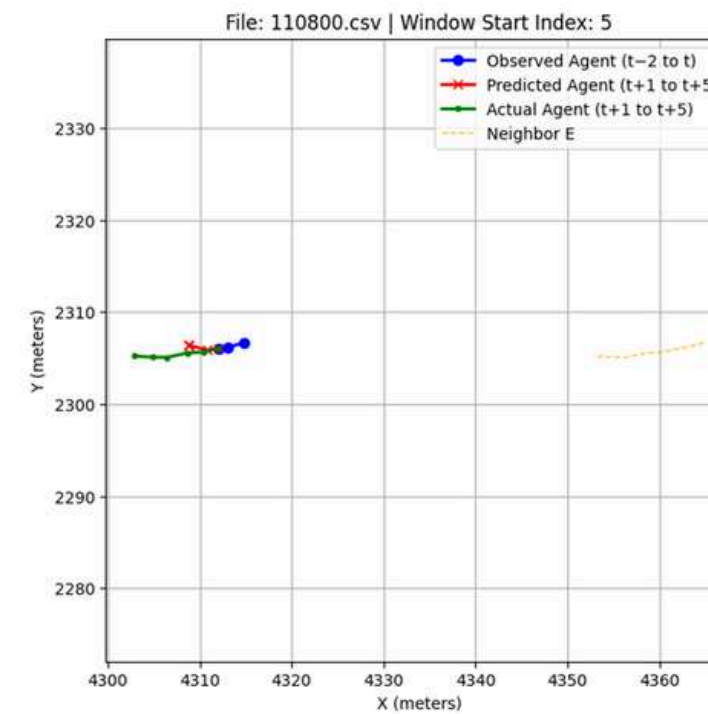


Fig-09: Predicted Trajectories after 20th Epoch

Result Analysis(Cont'd)

Predicted Trajectory of Implemented Model

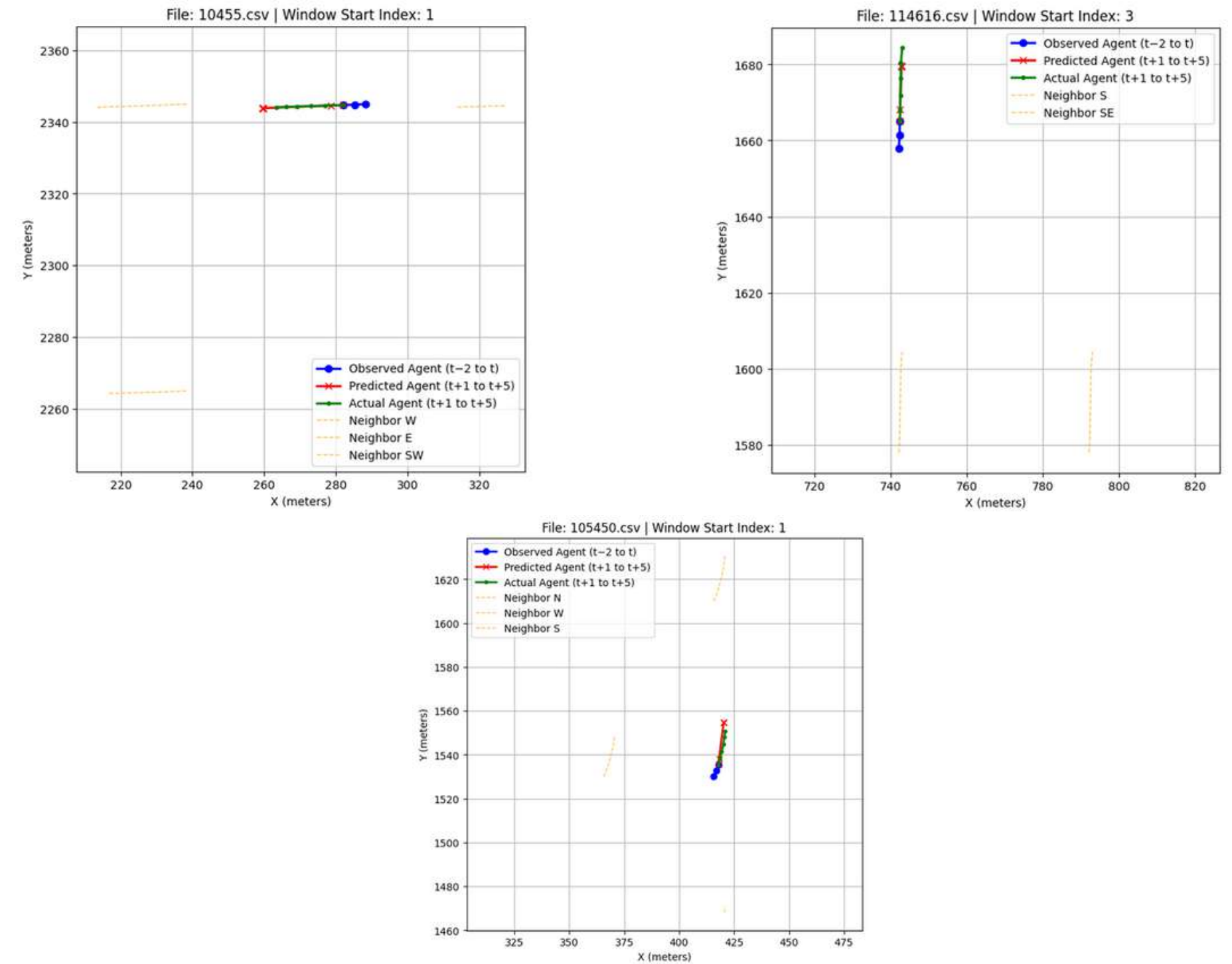


Fig-10: Predicted Trajectories after 25th Epoch

Result Analysis(Cont'd)

Evaluations: Comparison with Other State of the Art

Table 6: RMSE* Comparison of Proposed Model with Other Models

Horizon (s)	Dual Transformer[6]	GNN-RNN based Encoder-Decoder network[5]	BAT Model[4]	Intension-Aware Model [8]	HLTP Model [9]	Ours
1	1.16	0.67	0.23	0.42	0.39	0.55
2	1.76	0.99	0.82	1.01	0.91	0.68
3	2.43	1.21	1.54	1.69	1.45	1.07
4	3.12	1.55	-	2.56	2.09	1.53
5	3.64	2.14	-	3.67	2.93	1.92

* A low RMSE value indicates that the observed and actual data are close to each other and less prone to error.



Conclusion

Contribution

- Largest and well-defined dataset.
- Less complexity for window creation.
- Region based tracking.
- Position based on Timestamp.

Future Scope

- Use heavy dataset.
- Distillation Model.
- Issues with short term prediction.
- Multimodal model can be introduced.



References

1. NovAtel, “Sensor fusion and its role in safe autonomy.” Velocity Magazine (Tech Talk feature, NovAtel website), 2023. Accessed 4 June 2025.
2. Huang Y, Du J, Yang Z, Zhou Z, Zhang L, Chen H. A survey on trajectory-prediction methods for autonomous driving. *IEEE Trans Intell Veh.* 2022;7(3):652–74.
3. Chandra R, Bhattacharya U, Bera A, Manocha D. Traphic: Trajectory prediction in dense and heterogeneous traffic using weighted interactions. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition.* 2019. p. 8483–92.
4. Liao H, Li Z, Shen H, Zeng W, Liao D, Li G, et al. BAT: Behavior-aware human-like trajectory prediction for autonomous driving. In: *Proceedings of the AAAI Conference on Artificial Intelligence.* 2024;38:10332–40.
5. N. Deo and M. M. Trivedi, “Convolutional social pooling for vehicle trajectory prediction,” in *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pp. 1468–1476, 2018.
6. Gao K, Li X, Chen B, Hu L, Liu J, Du R, et al. Dual transformer-based prediction for lane change intentions and trajectories in mixed traffic environment. *IEEE Trans Intell Transp Syst.* 2023;24(6):6203–16.



References

7. M.-F. Chang, J. Lambert, P. Sangkloy, J. Singh, S. Bak, A. Hartnett, D. Wang, P. Carr, S. Lucey, D. Ramanan, and J. Hays, “Argoverse: 3d tracking and forecasting with rich maps,” in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern 77 Recognition (CVPR), pp. 8748–8757, 2019. Dataset URL: <https://www.argoverse.org>.
8. X. Chen, H. Zhang, F. Zhao, Y. Hu, C. Tan, and J. Yang, “Intention-aware vehicle trajectory prediction based on spatial-temporal dynamic attention network for internet of vehicles,” IEEE Transactions on Intelligent Transportation Systems, vol. 23, no. 10, pp. 19471–19483, 2022.
9. H. Liao, Y. Li, Z. Li, C. Wang, Z. Cui, S. E. Li, and C. Xu, “A cognitive-based trajectory prediction approach for autonomous driving,” IEEE Transactions on Intelligent Vehicles, 2024



Thank You

For your attention.

You may convey your thoughts and queries

